1. Observe

Ten unlabeled target observations

Target domain (unlabeled)



Target env. 1



Target env. 2



Target env. 3



Source reference (CORSET), cached offline: per-region features and statistics

2. Diagnose

Per-region adaptation cost, no fine-tuning

Pre-trained VLA (e.g., OpenVLA-OFT), partitioned into adapter regions

SigLIP Vision (V)	**DINOv2** Vision (V)	**Projector** Cross-modal (P)	**LLaMA** Language (L)	**Action head** Parallel L1 (A)

Tier-1: reference-based

Compare target vs. cached source features

$1 - \mathrm{CKA}(r)$, $\mathsf{F}(r)$

Tier-2: reference-free

Signals from the target data alone

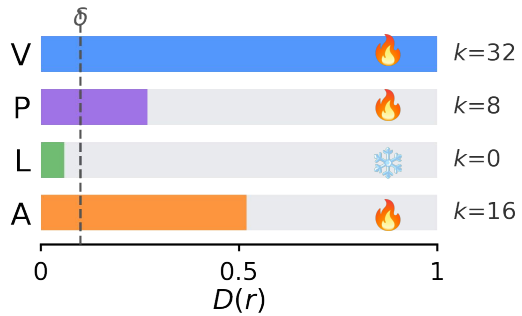
$\mathsf{G}(r)$, $\hat{\mathsf{U}}(r)$

Fréchet-distance gate $\mathsf{w}(r)$ blends the tiers → adaptation cost $\mathsf{D}(r)$ per region (Fig. 2)

3. Allocate

Variable-rank LoRA adapters

Rank $\mathsf{k}(r) \propto \mathsf{D}(r)$; frozen if $\mathsf{D}(r) < \delta$



❄ Frozen 🔥 Trainable

Budget: 0.04% – 2% of model parameters trainable

4. Adapt

LoRA fine-tuning with allocated ranks



Adapted VLA on the xArm-7

Train only the allocated adapters; all other parameters frozen